

Use case	Key features	Future trends
Gaming and entertainment	Highly realistic graphics and physics simulation	Integration of haptic feedback for enhanced immersion
	Immersive audio and spatial sound	Seamless integration of virtual and augmented reality
	Interactive gameplay and lifelike characters	Real-time multiplayer experiences across platforms
Architecture and design	Detailed material simulations	Integration of AI for automated design optimisations
	Accurate lighting effects	Real-time collaboration and feedback during design process
	Virtual walkthroughs and spatial interactions	Integration with Building Information Modelling (BIM)
Education and training	Realistic simulations of various scenarios	Adaptive learning experiences based on user interactions
	Hands-on training in safe environments	Integration of natural language processing for dialogue
	Experiential and immersive learning	Virtual mentorship and guidance for skill development
Retail and e-Commerce	Immersive 3D virtual stores	Personalised recommendations based on user preferences
	Accurate product visualisation	Integration of virtual try-on technologies
	Interactive product interaction	Integration with blockchain for secure transactions
Healthcare and telemedicine	Virtual consultations in realistic environments	Real-time patient monitoring and vital sign integration
	Medical simulations for training and practice	Integration with medical imaging technologies
	Enhanced visualisation for accurate diagnosis	Integration of telepresence technologies for remote surgery

Table 1: Use cases, features and future trends of hyper-real platforms

Gaming and entertainment: Hyper-real platforms have the potential to revolutionise the gaming and entertainment industry within the metaverse. By leveraging advanced graphics rendering, realistic physics simulations, and immersive audio, these platforms can create breathtaking virtual worlds and experiences. Players can step into hyper-realistic environments, interact with lifelike characters, and engage in highly immersive gameplay. From virtual reality (VR) gaming to augmented reality (AR) experiences, hyper-real platforms offer unprecedented levels of immersion, enhancing entertainment experiences and captivating audiences like never before.

Industrial architecture and design: In the realm of architecture and design (e.g., automobile industrial design), hyper-real platforms can reshape the way professionals visualise and present their ideas. Architects and designers can

create virtual models of buildings and spaces, offering clients an immersive walkthrough experience that closely resembles reality. Hyper-real platforms enable detailed material simulations, accurate lighting effects, and realistic spatial interactions, empowering stakeholders to make informed decisions during the design process. This immersive approach not only enhances collaboration but also saves time and resources by identifying potential issues before construction begins.

Education and training: Hyper-real platforms hold significant promise for education and training applications within the metaverse. By creating realistic simulations of various scenarios, such as medical procedures, hazardous environments, or historical events, these platforms can provide immersive and experiential learning opportunities. Students and

professionals can engage in hands-on training, practice skills in a safe environment, and gain practical knowledge that translates into real-world scenarios. The hyper-realistic nature of these platforms enhances engagement and knowledge retention, transforming the way we learn and train across diverse fields.

Retail and e-commerce: Hyper-real platforms offer exciting opportunities for the retail and e-commerce sectors in the metaverse. Virtual stores can replicate the physical shopping experience by providing immersive 3D environments where customers can browse and interact with virtual products. Hyper-realistic visuals and accurate representations enable users to inspect products in detail, try on virtual clothing, or even simulate the placement of furniture in their homes. This seamless integration of virtual